HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed synthetic workload queries and datasets on **AWS S3** to mirror production environments for testing DataPelago Accelerator for Spark (DPA-S), reducing test setup time by **30%**.
- Built analysis tools using **Python** and **SQL** to parse and correlate logs from DPA-S, Apache Spark, and cloud environments, decreasing average issue diagnosis time by **40%**.
- Participated in a rotating product support schedule, addressing and escalating customer issues related to data processing anomalies and performance bottlenecks.
- Conducted end-to-end product validation for DPA-S, collaborating with engineering teams to define and execute test plans across structured and unstructured data scenarios.
- Analyzed artifacts from product validation and customer-raised issues to identify configuration tuning opportunities and potential product bugs, contributing to **15+** defect reports.
- Created comprehensive bug reports, Root Cause Analysis (RCA) documents, and configuration guides that improved customer onboarding efficiency for new DPA-S deployments.
- Containerized DPA-S components using **Docker** and deployed them to an **AWS EKS** cluster, ensuring consistent environments for testing and customer deployments.

SKILLS

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch, EKS), GCP

Data Processing Frameworks – Apache Spark, Apache Flink, Ray

Languages & Scripting – Python, SQL, Java, Shell Scripting

Databases – PostgreSQL, MySQL, Oracle

Tools & Technologies – Docker, Kubernetes, Jenkins, Git, Jira, Confluence, Splunk, Dynatrace

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, R, pandas, NumPy, scikit-learn

PROJECTS

Data Processing Engine Validation Suite (DataPelago)

- Developed a suite of synthetic datasets and workload queries using **Python** and **SQL** to simulate diverse customer data patterns (structured, unstructured) for testing the DataPelago Nucleus engine.
- Created diagnostic scripts to analyze logs from **Apache Spark** and the DataPelago engine, identifying performance regressions and functional anomalies, which were then used for proactive bug fixing.
- Implemented automated validation checks against output schemas and data integrity metrics, improving test coverage by **20%** and reducing manual verification effort.

Customer Support Automation Tools (DataPelago)

- Engineered a Python-based tool to parse and summarize logs from distributed data processing environments, enabling faster root cause analysis for customer-reported issues.
- Integrated the tool with a ticketing system workflow, automatically flagging critical errors and providing initial diagnostic summaries to the support team, reducing response time by approximately **15%**.
- Developed documentation and best practice guides for customers on log analysis and common troubleshooting steps, directly contributing to improved customer self-sufficiency.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)